Turnout variables: symbol, code, transform, source

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A reference map for the turnout side of the precinct × vote-mode model, tracing each quantity from the raw data column it comes from to the outcome that is actually fitted. Everything here assumes the `USE_COMPOSE = 1` default. Line numbers refer to `scripts/model/tx_gov_trial.R` unless another file is named.

1 Inputs

Sym.	Code	Transform	Source column → dataset
n_i	`n`	passthrough	`n` → `TX_precinct.parquet`. The YouGov frame count; L2's is kept separately as `l2_n`
—	`turnoutPred`	passthrough	`turnoutPred` → `yougov_precinct_frame.parquet`. A vendor midterm turnout score; not computed by us
E_i	`est_turnout`	$\min(\texttt{turnoutPred} \cdot n_i,\ n_i)$	`prep_tx_model_inputs.R:319`
E_i	`est_tot`	`est_dem + est_rep`, re-widened from the long-by-party frame	`est` → `TX_precinct.parquet` via `covs_long`; line 703
v_{im}^{hist}	`voters`	passthrough	`vote22_{mail,early,eday}` → L2, pivoted at `prep_tx_model_inputs.R:363--370` → `TX_precinct_mode_history.parquet`

2 Derived priors

Sym.	Code	Transform	Source
p_{im}^0	`mode_turn_share`	$\frac{v_{im}^{\text{hist}} + k\, c_{cm}}{\sum_{m'} v_{im'}^{\text{hist}} + k}$, $k = 40$, then renormalised to sum to 1	`voters` → mode history; lines 645–653. c_{cm} is the county's mode share
τ_{im}^0	`prior_tau`	$E_i \cdot p_{im}^0$	line 745
κn_i	`cap`	$1.25 \cdot \max(n_i,\ 1)$	line 735
θ_i^0	`y_tau_prior` *(ref slot)*	$\text{logit}(E_i/\kappa n_i)$	lines 762–764
η_{im}^0	`y_tau_prior` *(other slots)*	$\log\!\left(p_{im}^0 / p_{i,\text{ref}}^0\right)$	line 765

3 Observed, from the returns

Sym.	Code	Transform	Source
c_{im}	`tau`	`votes_dem + votes_rep + votes_oth`	the live feed, or `TX_truth.parquet` + `TX_schedule.parquet` in a sim; line 894
—	`obs`	`stage >= 2 & tau > 0 & tot2 > 0`	line 896. "has reported"
—	`obs_turn`	`obs & (complete | tau >= 1.1 * prior_tau)`	line 918. "has finished". The fallback is the plateau proxy `COMPLETE_FRAC`
C_i	`cnt_prec`	$\sum_m c_{im}$ over `obs` modes	line 915
T_i^{fin}	`T_final`	$\sum_m c_{im}$ over `obs_turn` modes	line 917

4 Fitted outcomes — the same three slots, two different meanings

The per-mode turnout slot is *reinterpreted* rather than restructured, so that Σ, the error draws, the conditional-MVN grouping and the aggregation offsets all keep working unchanged. The reference mode's slot carries the precinct total; the other two carry the mode composition.

Sym.	Code	Transform	Fit condition
θ_i	`y_tau` *(ref slot)*	$\text{logit}\big(T_i^{\text{fin}}/\kappa n_i\big)$	`all_done` — *every* mode of the precinct finished
η_{im}	`y_tau` *(other slots)*	$\log\big(c_{im}/c_{i,\text{ref}}\big)$	`obs_turn & ref_done` — this mode *and* the reference
—	`lb_tau`	$\text{logit}(C_i/\kappa n_i)$	censoring bound; reference slot only, and only when `USE_CENSOR=1`

5 Back out to votes

Sym.	Code	Transform
T_i	`T_draw`	$\kappa n_i \cdot \text{logit}^{-1}(\theta_i + \varepsilon)$
p_{im}	—	$\text{softmax}(\eta_i)$, with the reference mode's log-ratio fixed at 0
τ_{im}	`tau` (draw)	$T_i \cdot p_{im}$ — sums to T_i exactly, and $T_i < \kappa n_i$ by construction
—	—	$T_i \leftarrow \max(T_i,\ C_i)$ — the counted-votes floor

6 Switches that are off by default

Everything above describes the shipped defaults. The variables below exist in `tx_gov_trial.R` but do nothing unless their switch is set, which is why they do not appear in the tables above. They are the machinery for letting a *partially* counted precinct inform θ.

Sym.	Code	Transform	Switch / status
δ_m	`comp_tilt`	vote-weighted mean of $\log(c_{im}/c_{i,\text{ref}}) - \eta^0_{im}$ over precincts where mode m *and* the reference are both counted out; $\delta_{\text{ref}} \equiv 0$	`COMPOSITION_UPDATE=1`. Estimates a statewide shift in the mode mix
p_{im}	`p_tilt`	$p^0_{im} \exp(\delta_m)$, renormalised per precinct	equals p^0_{im} when the tilt is off
q_i	`q_cov`	$\sum_{m \text{ counted out}} p_{im}$ — the share of the precinct's expected vote that is finished. `all_done` $\Rightarrow q_i = 1$	always computed; only *used* by the switches below
$\widehat{T}_i$	`T_hat`	T_i^{fin} when `all_done`, else T_i^{fin}/q_i	`MIN_MODE_COVERAGE` < 1. The gross-up
—	`w_tau`	$q_i^{\texttt{COVERAGE_WT_POW}}$, default exponent 2	precision weight: $\text{Var}(\widehat{T}) \propto 1/q^2$
r_i	`r_prior`	$C_i / \sum_m \tau^0_{im}$	counted votes over the prior expected total. Needs no completeness flag
π_i	`p_atom`	clamp(r_i, 0.02, 0.98) under the default `ATOM_SOURCE=prior`; `flag` uses q_i and `both` their max	`USE_ATOM=1`. P(precinct has finished), the point mass at the bound

Why θ's gate is the one that is wrong. A completed cell's per-mode ratio decomposes exactly:

$$\log\left(\frac{c_{im}}{\tau^0_{im}}\right) = \underbrace{\log\left(\frac{T_i}{E_i}\right)}_{\text{level } (\theta)} + \underbrace{\log\left(\frac{p_{im}}{p^0_{im}}\right)}_{\text{composition } (\eta)} .$$

One completed mode identifies the level given a composition; *two* identify both. So η's condition (this mode and the reference) is already correct, and only θ's `all_done` is too strict — it demands three modes where one suffices.

Measured status, so none of this reads as settled. On a simulated night whose true turnout was 0.85× the prior, relaxing θ's gate to $q_i \geq 0.5$ moved turnout from 119% of truth to 101%, and did not degrade the correctly-specified night. But the *same* setting on the real TX 2024 backtest, where completeness is a plateau proxy rather than a feed flag, fabricated votes — 103.7% → 109.0% at full count — because $1/q$ invents the share of any mode that is finished but not flagged. Censoring (`USE_CENSOR=1`), with or without the atom, made turnout *worse* in both simulated scenarios: a lower bound can only push the estimate up, and the error it was aimed at points down. All of these default to off for that reason.

Two things the table makes visible

`est_turnout` **enters three times.** As E_i it sets the θ prior (line 764); multiplied by the mode split it becomes `prior_tau` (line 745); and `prior_tau` is then the denominator of the `COMPLETE_FRAC` plateau test (line 918). So a single vendor score sets the turnout *level*, the mode *composition*, and the *completeness proxy* at once. It is the most load-bearing number on the turnout side, and none of it is computed here.

The fit-condition column is where the model breaks. θ and η occupy the same three slots and look parallel everywhere in the code — same variable name `y_tau`, same `prior_tau` ancestry, same loop. But θ requires *all three* modes to be finished and η requires *two*. That asymmetry is correct as identification — one finished mode pins the level only if the composition is assumed, two pin both — but nothing in the naming signals it. Since mail votes finish days after election night, `all_done` is effectively never satisfied while results are being reported, the θ equation is fitted on zero rows, and

the solve returns its prior unchanged. Measured on a simulated night whose true turnout was $0.85\times$ the prior, the model reported 119% of truth from the first checkpoint to the last — which is $1/0.85$, the prior echoed back.